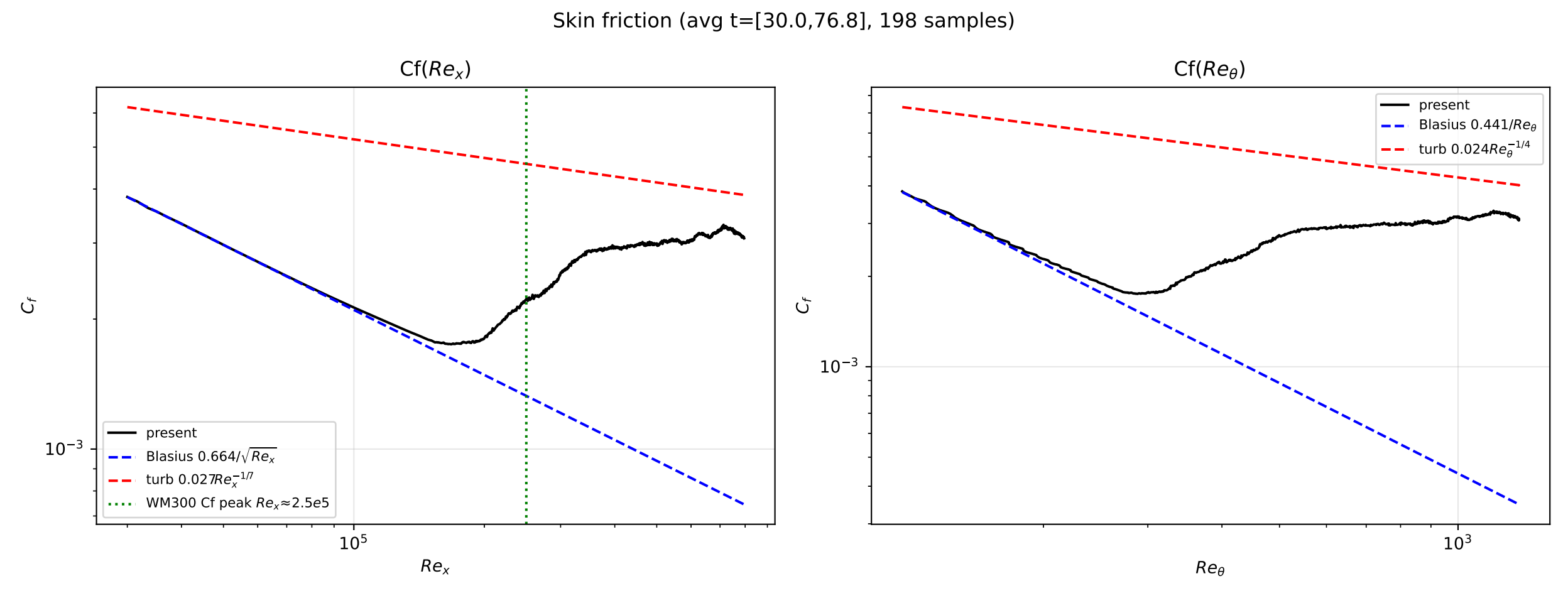
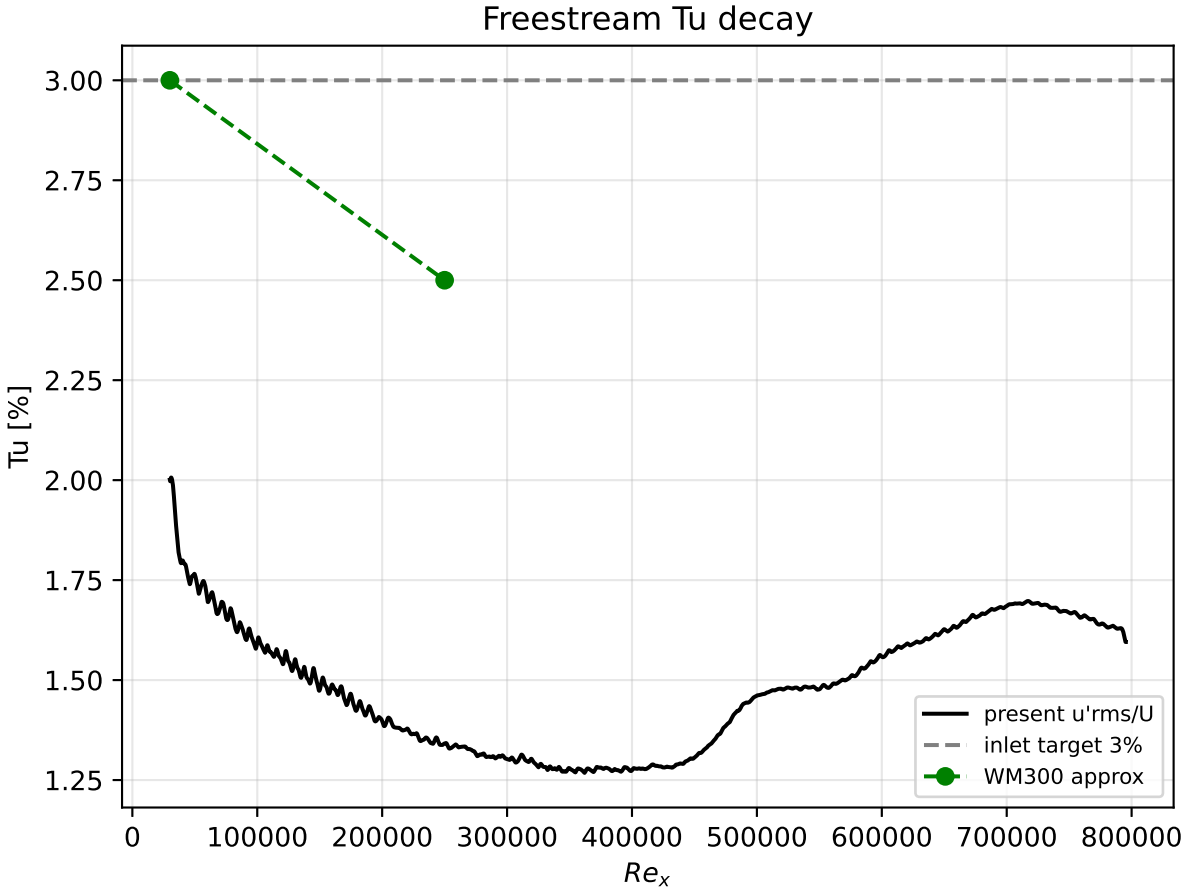
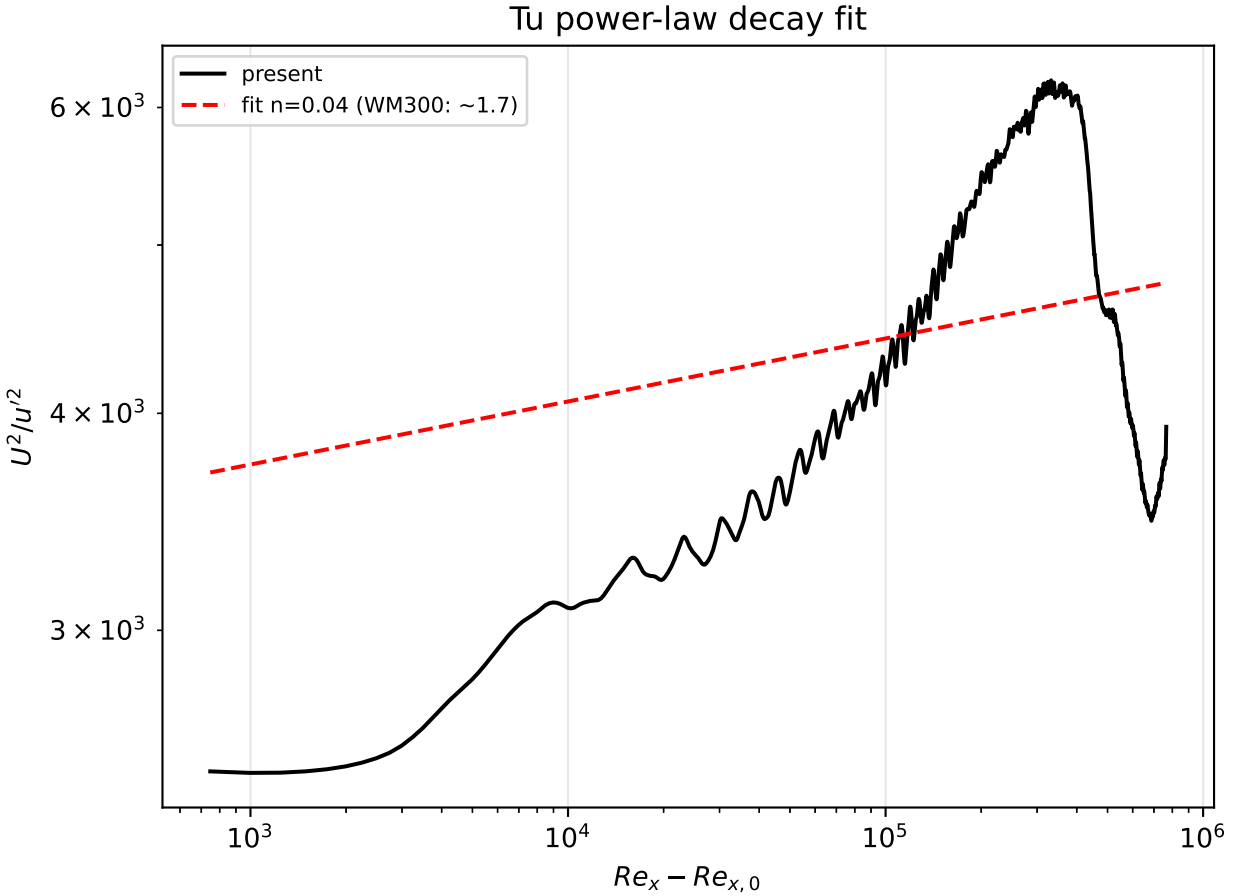
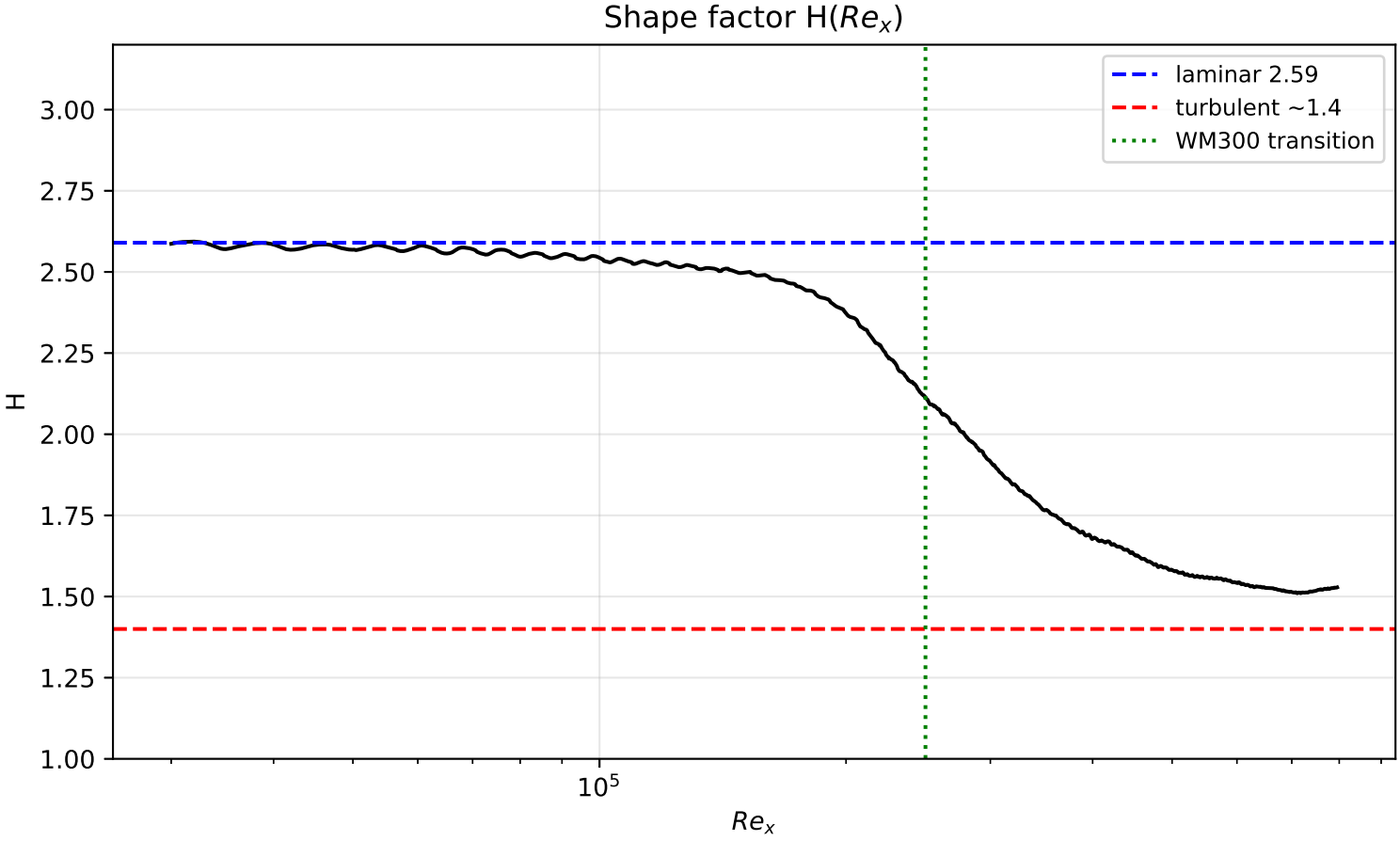
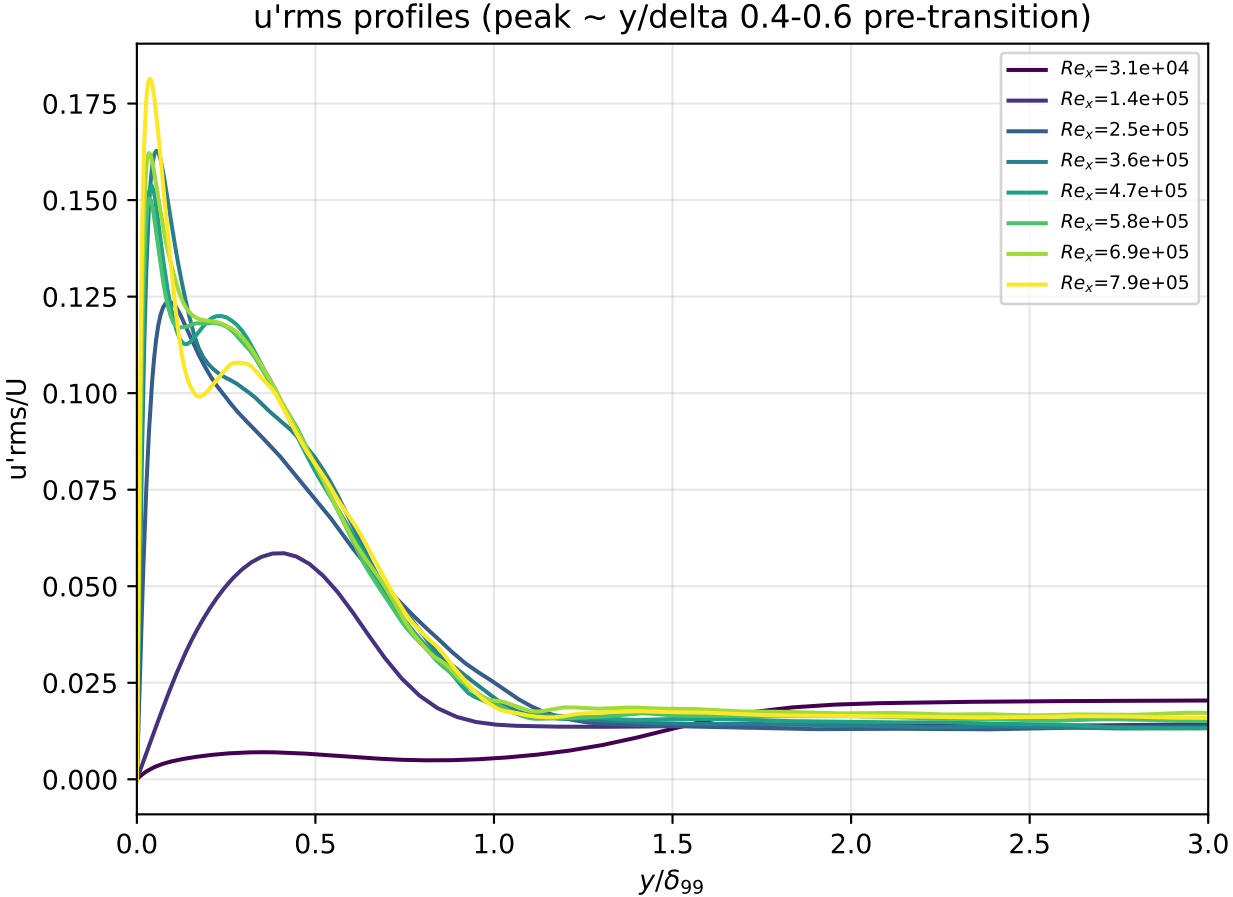
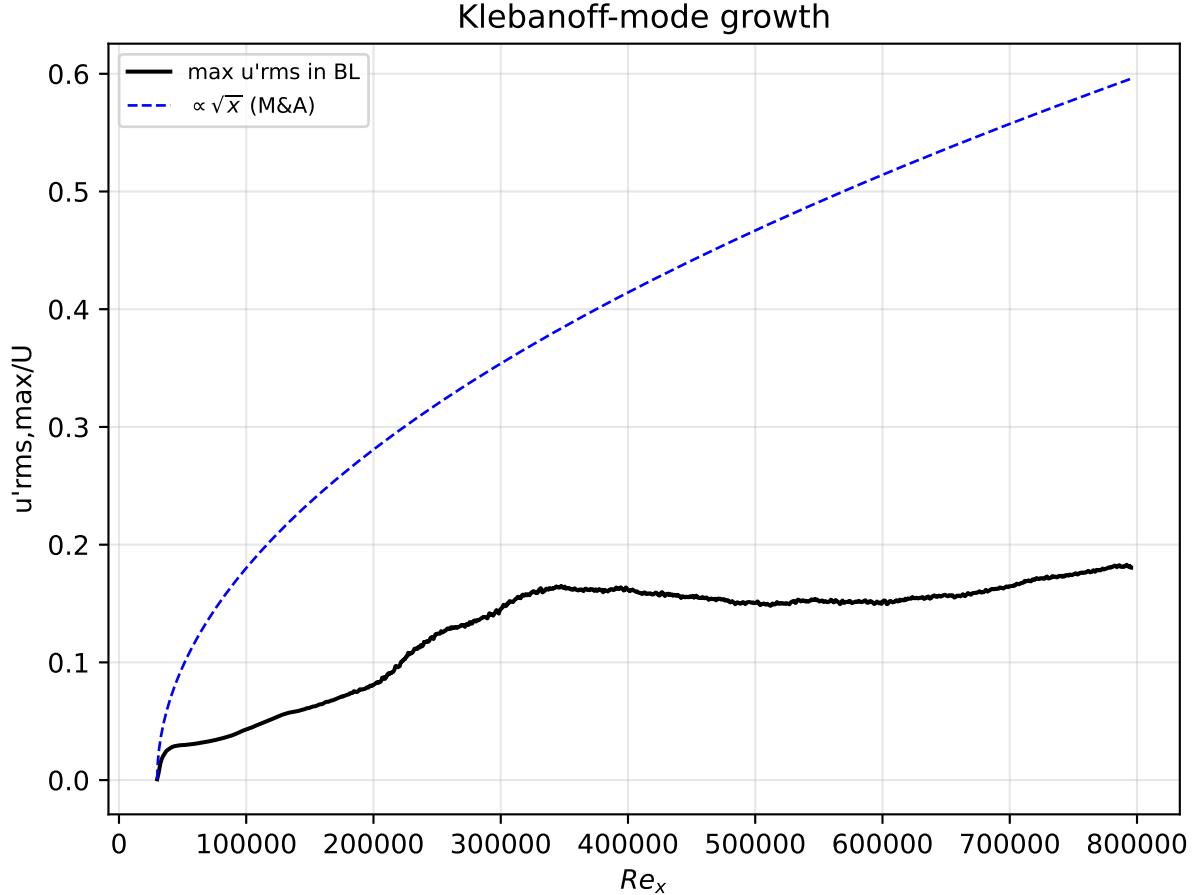

Skin friction (avg t=[30.0,76.8], 198 samples)
Cf(Re_x)
Cf(Re_θ)
C_f
10^{-3}
10^5
10^3
Re_x
Re_θ
present
Blasius $0.664/\sqrt{Re_x}$
turb $0.027Re_x^{-1/7}$
WM300 Cf peak $Re_x \approx 2.5e5$
Blasius $0.441/Re_\theta$
turb $0.024Re_\theta^{-1/4}$

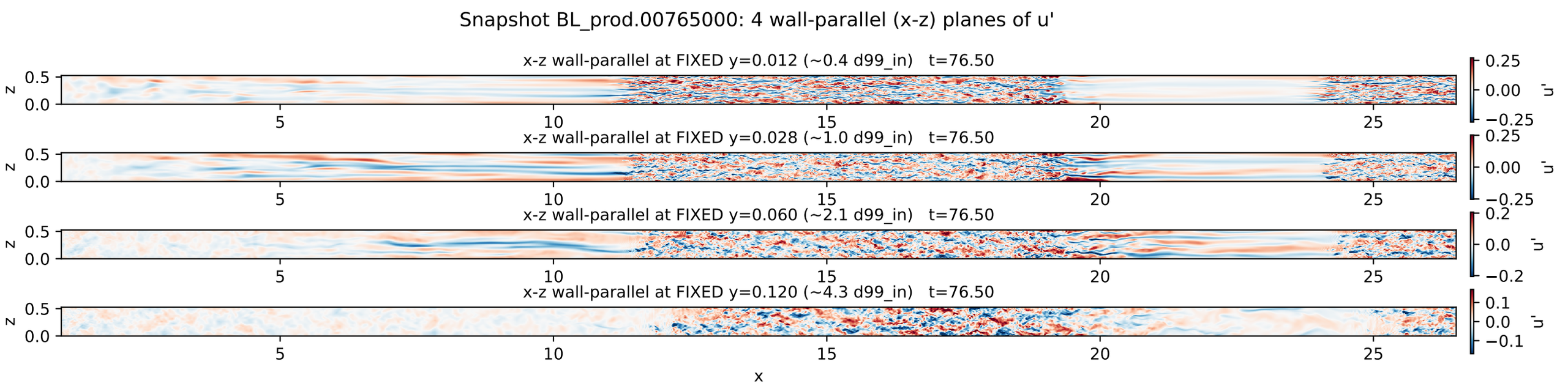

Freestream Tu decay
Tu [%]
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
0
100000
200000
300000
400000
500000
600000
700000
800000
Re_x
present u'rms/U
inlet target 3%
WM300 approx

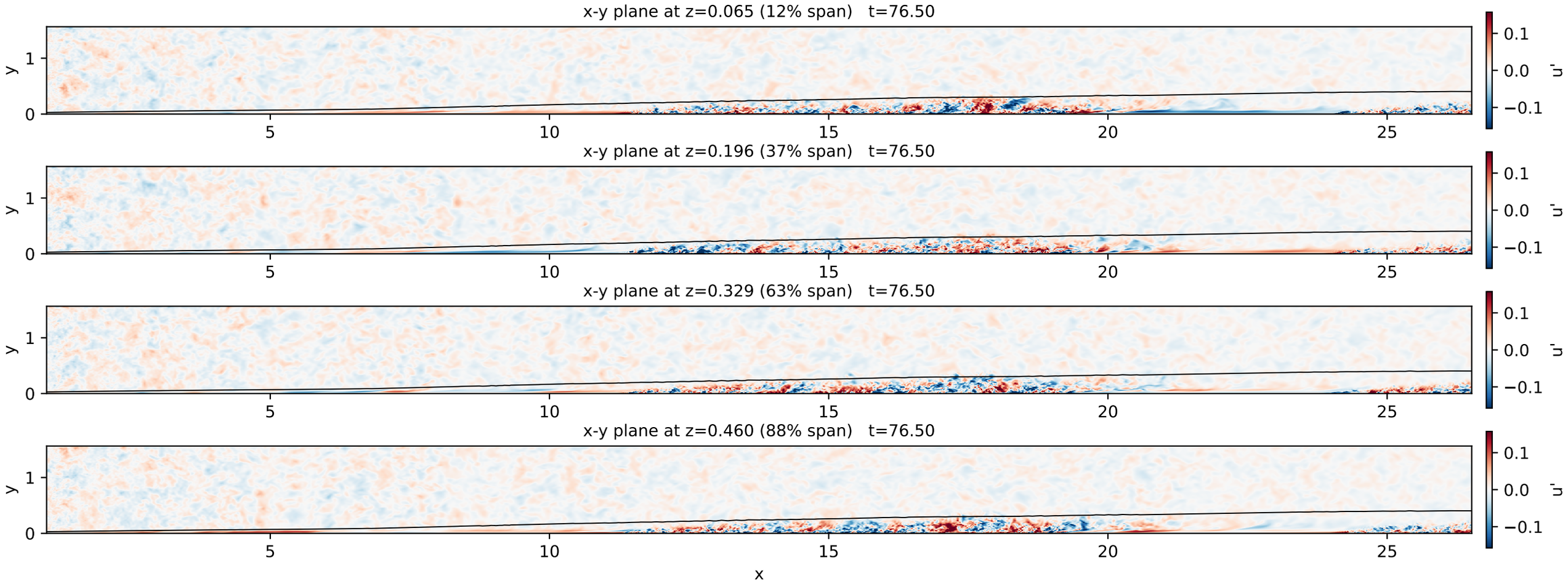

Tu power-law decay fit
present
fit n=0.04 (WM300: ~1.7)
6×10^3
4×10^3
3×10^3
U^2/u'^2
10^3
10^4
10^5
10^6
$Re_x - Re_{x,0}$


Shape factor H(Re_x)
laminar 2.59
turbulent ~1.4
WM300 transition
H
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
10^5
Re_x


u'rms profiles (peak ~ y/delta 0.4-0.6 pre-transition)
Re_x=3.1e+04
Re_x=1.4e+05
Re_x=2.5e+05
Re_x=3.6e+05
Re_x=4.7e+05
Re_x=5.8e+05
Re_x=6.9e+05
Re_x=7.9e+05
u'rms/U
y/δ99
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.0
0.5
1.0
1.5
2.0
2.5
3.0


Klebanoff-mode growth
max u'rms in BL
∝ √x (M&A)
u'rms,max/U
Re_x
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
100000
200000
300000
400000
500000
600000
700000
800000


Snapshot BL_prod.00765000: 4 wall-parallel (x-z) planes of u'
x-z wall-parallel at FIXED y=0.012 (~0.4 d99_in) t=76.50
x-z wall-parallel at FIXED y=0.028 (~1.0 d99_in) t=76.50
x-z wall-parallel at FIXED y=0.060 (~2.1 d99_in) t=76.50
x-z wall-parallel at FIXED y=0.120 (~4.3 d99_in) t=76.50
z
x
u'
0.5
0.0
5
10
15
20
25
0.25
0.00
−0.25
0.2
−0.2
0.1
−0.1


Snapshot BL_prod.00765000: 4 (x-y) planes of u' at different z
x-y plane at z=0.065 (12% span) t=76.50
x-y plane at z=0.196 (37% span) t=76.50
x-y plane at z=0.329 (63% span) t=76.50
x-y plane at z=0.460 (88% span) t=76.50
y
x
u'
0
1
5
10
15
20
25
0.1
0.0
−0.1